

Foundation (Jalapeno)

- Q1.** Which lines are parallel if they have the same m in $y = mx + b$?
- (A) Slopes of 2 and 2
(B) Slopes of 3 and -3
(C) Slopes of 1 and 1
(D) Y-intercepts of 4 and 4
(E) Y-intercepts of 5 and -5
- Q2.** Which of the following line equations has a slope of 2?
- (A) $y = x + 1$
(B) $y = 2x - 3$
(C) $y = x - 4$
(D) $y = -2x + 1$
(E) $y = 3x + 2$
- Q3.** If two lines have slopes of m_1 and m_2 , they are parallel if $m_1 =$
- (A) $m_2 - 1$
(B) m_2
(C) $m_2 + 1$
(D) $-m_2$
(E) m_2^2
- Q4.** What is the equation of a line parallel to $y = 3x - 2$ that passes through $(1, 4)$?
- (A) $y = 3x + 1$
(B) $y = 2x - 1$
(C) $y = 3x - 1$
(D) $y = 4x - 3$
(E) $y = 3x + 2$
- Q5.** In a space mission, a spaceship travels at a slope of 2. If a parallel spaceship has the equation $y = 2x + 5$, the first spaceship's equation is
- (A) $y = 2x - 3$
(B) $y = x + 2$
(C) $y = 2x - 1$
(D) $y = x - 4$
(E) $y = 3x + 1$
- Q6.** If two lines are parallel and have equations $y = mx + b_1$ and $y = mx + b_2$, then b_1 must equal
- (A) $b_2 - 1$
(B) b_2
(C) $b_2 + 1$
(D) $-b_2$
(E) b_2^2
- Q7.** What is the relationship between the slopes of parallel lines in the context of planetary orbits?
- (A) They are always equal to the gravitational constant
(B) They are always less than the speed of light
(C) They are always equal
(D) They are always perpendicular
(E) They are always unequal
- Q8.** Which of the following is an example of parallel lines in a lab experiment?
- (A) A microscope and a telescope
(B) A test tube and a beaker
(C) Two rows of petri dishes
(D) A Bunsen burner and a hot plate
(E) A thermometer and a barometer

Open Ended Questions (FRQ)

- Q9.** A space probe is traveling at a slope of 4. If a parallel probe has the equation $y = 4x - 2$, write the equation of the first probe's path in the format $y = mx + b$. Assume the first probe passes through $(0, 1)$.
- Q10.** In a physics experiment, two lines are parallel and have equations $y = 2x + 1$ and $y = 2x - 3$. Show that these lines have the same slope and find the y -intercept of a third parallel line that passes through $(2, 5)$. Use $y = mx + b$ format.

Chef's Tips

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- Tip 1: Ensure students understand the concept of parallel lines and slopes before administering this test.
- Tip 2: Emphasize the importance of accurate calculations and units in space and science applications.
- Tip 3: Allow students to use graph paper and calculators to help visualize and solve problems.